School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Milestone-3 Scribe

S2_G11_CLI

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Problem Statement

Deterministic extrapolation of atmospheric conditions fails rapidly due to the chaotic nature of fluid dynamics. **Quantitative Precipitation Nowcasting (QPN)** attempts to forecast rainfall intensity over very short time horizons (0–6 hours) using radar observations. Weather systems evolve according to highly non-linear atmospheric dynamics; even small uncertainties in initial conditions lead to rapidly diverging forecasts, making purely deterministic prediction mathematically ill-posed for lead times beyond a few hours.

Scribe Question 1: System Objective and Deterministic Baseline

System Objective

The objective of the system is to forecast the spatial rainfall intensity field $R(x,t)$ over a short time horizon. Given a sequence of discrete radar observations $\mathbf{Z}_{1:t}$ up to time t, the system must estimate the predictive distribution of future rainfall:

$$P(R(x,\, t+\tau) \mid \mathcal{Z}_{1:t}) \tag{1}$$

for lead times $\tau \in [0,\, 6\text{ h}]$. In Milestone 3 we implement the **deterministic point forecast** — the Lagrangian Persistence model — which replaces this full distribution with a single best-estimate trajectory.

Deterministic Baseline: Lagrangian Persistence

The deterministic baseline assumes that rainfall patterns are transported by the atmospheric flow *without change in intensity.* This is the **Lagrangian Persistence** assumption (Zawadzki et al., 1994) — the standard baseline in operational nowcasting. It is physically motivated by the dominance of advection over growth/decay for very short lead times (0–60 min), and outperforms NWP models in this window (Berenguer et al., 2012).

Let $V(x,t) = (u,v)$ be the advection velocity field. The conservation equation for incompressible flow is the advection PDE:

$$\frac{\partial R}{\partial t} + V(x,t) \cdot \nabla R = 0 \tag{2}$$

Setting $dR/dt = 0$ (no sources or sinks) and discretising with $\Delta t = 5$ minutes, the **deterministic forecast update rule** is:

$$R(x,\, t+\Delta t) = R(x - V(x,t)\,\Delta t,\; t) \tag{3}$$

This rule is applied iteratively for $\tau = \Delta t,\, 2\Delta t,\, \ldots,\, 72\Delta t$ to produce a 6-hour forecast. The velocity field $V(x,t)$ is estimated at time t and held **frozen** (constant) over the forecast horizon — a key simplification examined in Question 6.

Why this baseline is reasonable and commonly used: Lagrangian persistence is the industry-standard baseline in operational nowcasting. It requires no assumptions about thermodynamic processes, is computationally trivial, and in the pysteps library's semi-Lagrangian implementation (`pysteps: extrapolation.semilagrangian`) avoids numerical diffusion by tracing parcel trajectories and interpolating only at the final step (Germann and Zawadzki, 2002). The pysteps paper confirms it achieves CSI above 0.8 for the first 60–90 minutes on MeteoSwiss and FMI datasets.

Scribe Question 2: Random Variables in the Simulation

Although the baseline forecast rule is fully deterministic, the system inputs originate from inherently random physical processes. Three foundational random fields govern the state-space model established in Milestones 1 and 2.

State Variable: Rainfall Intensity $R(x,t)$

Formal definition: $R(x,t)$ is a non-negative continuous random field, $R : \Omega \times \mathcal{T} \to \mathbb{R}_{\geq 0}$, representing precipitation at spatial location x and time t.

Assumed distribution: Empirical observations show that rainfall is highly positively skewed. Accordingly:

$$R(x,t) \;\sim\; \text{LogNormal}(\mu,\, \sigma^2) \tag{4}$$

Transformation for tractability (`pysteps: utils.transformation`): To enable linear Gaussian computations, the dBR transformation is applied with explicit thresholding to handle zero-inflation:

$$X = 10 \cdot \log_{10}(R) \quad \text{for } R \geq 0.1 \text{ mm/h} \qquad \Bigg| \qquad X = -15 \text{ dBR} \quad \text{for } R < 0.1 \text{ mm/h}$$

The value of -15 dBR assigns 0.03 mm/h to dry pixels, preventing logarithmic singularities. Under this transformation, the latent state $X(x,t) = 10 \log_{10} R$ is approximately:

$$X(x,t) \;\sim\; \mathcal{N}(\mu_X,\, \sigma_X^2) \tag{5}$$

Role in deterministic baseline: $X(x,t)$ is the quantity that is advected by the semi-Lagrangian scheme. The Gaussian structure allows linear interpolation without distorting statistical properties. The final forecast is inverted via $\hat{R} = 10^{X/10}$.

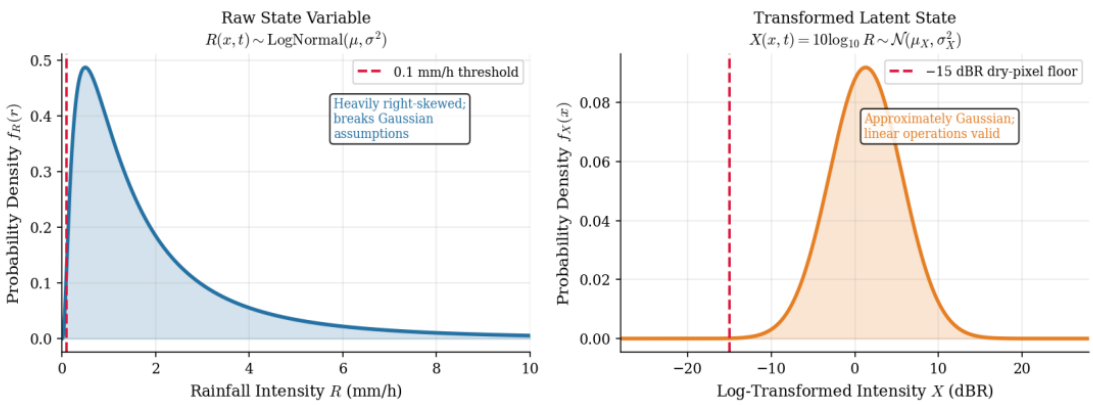

Figure 1: Probabilistic Transformation of State Variable $R(x,t)$ ***(pysteps: utils.transformation.dBR).*** *Left — Log-normal PDF of raw rainfall* $R(x,t)$*, showing the heavy right-skew that breaks Gaussian assumptions. Right — Gaussian PDF of the transformed latent state* $X(x,t) = 10 \cdot \log_{10}(R)$*. Dashed red line shows the* -15 *dBR dry-pixel floor used in pysteps.*

Forcing Variable: Velocity Field $V(x,t)$

Formal definition: $V(x,t) = (u,v) \in \mathbb{R}^2$ is a random vector field describing the motion of rainfall structures.

Assumed distribution: V is estimated via the optical flow equation (brightness constancy constraint) with a Gaussian perturbation accounting for estimation uncertainty:

$$V(x,t) = \mathbb{E}[V(x,t)] + \xi_V(x,t), \qquad \xi_V \sim \mathcal{N}(0,\, \Sigma_V) \tag{6}$$

Role in deterministic baseline: In Milestone 3 we use only the point estimate $\mathbb{E}[V(x,t)]$ from the **Lucas-Kanade** optical flow algorithm (`pysteps: motion.lucaskanade`), which solves the brightness constancy PDE with a spatial continuity regularisation. The perturbation ξ_V and the temporal evolution $\partial V/\partial t$ are both set to zero.

Innovation Variable: $N(x,t)$

Formal definition: $N(x,t)$ represents sub-grid thermodynamic uncertainty — spontaneous convective initiation, intensification, and decay that cannot be captured by advection.

Assumed distribution (from Milestone 2): Spatially correlated Gaussian random field:

$$N(\,\cdot\,,\, t) \;\sim\; \mathcal{GP}(0,\; K(x,\, x')) \tag{7}$$

where $K(x,x')$ is a positive-definite spatial covariance kernel with exponential form $\rho(h) = e^{-h/\lambda}$. **Role in deterministic baseline:** We set $N(x,t) = 0$ identically — the defining characteristic of the Milestone 3 deterministic rule.

Scribe Question 3: Probabilistic Modeling and Fixed Decision Rule

Governing Stochastic PDE (Full Probabilistic Model)

The full probabilistic model, derived in Milestone 2 from the Lagrangian material derivative, is governed by the Stochastic PDE:

$$\frac{\partial X}{\partial t} + V(x,t) \cdot \nabla X = N(x,t) \tag{8}$$

Discretising with the method of characteristics along advection trajectories yields the full stochastic state transition equation:

$$X(x,\, t+\Delta t) = X(x - V(x,t)\,\Delta t,\ t) + N(x,t) \tag{9}$$

Deterministic Simplification and Decision Rule

Setting $N(x,t) = 0$ in the full stochastic equation, the **deterministic decision rule** for Milestone 3 becomes:

$$\boxed{X(x,\, t+\Delta t) = X(x - V(x,t)\,\Delta t,\ t) \qquad [N(x,t) = 0]} \tag{10}$$

This is the Lagrangian Persistence forecast. It is crucial to distinguish what is and is not removed by this simplification. The inputs $R(x,t)$ and $V(x,t)$ *remain random* — they originate from the stochastic atmosphere and carry their full probabilistic structure. What the deterministic rule removes is the explicit injection of new uncertainty at each forecast step. The decision function is non-randomised; only the *inputs* are stochastic.

Probabilistic Assumptions Remaining Active in the Baseline

- **Log-normality of $R(x,t)$:** The raw rainfall field is still modelled as LogNormal(μ, σ^2). The dBR transformation that makes X approximately Gaussian is essential for the linear advection operator to preserve distributional properties. Without this, interpolation during advection would distort the statistical structure of the forecast fields.

- **Spatial correlation of the input field:** The correlation structure of $R(x,t)$ persists through the advection step. Adjacent grid points remain correlated because the same velocity field advects the spatially coherent input field together. The exponential spatial correlation function:

  $$\rho(h) = e^{-h/\lambda}, \qquad h = \|x - x'\|_2 \tag{11}$$

  is active in the input field, where λ is the decorrelation length scale (typically 10–40 km). This correlation structure ensures that short-lead-time forecast fields look physically realistic and have proper spatial power spectra.

- **Gaussian approximation of $X(x,t)$:** The latent state is modelled as a Gaussian process in transformed space. This ensures that the deterministic forecast preserves correct second-order statistics and justifies using RMSE as a metric and the Gaussian error model for residual analysis.

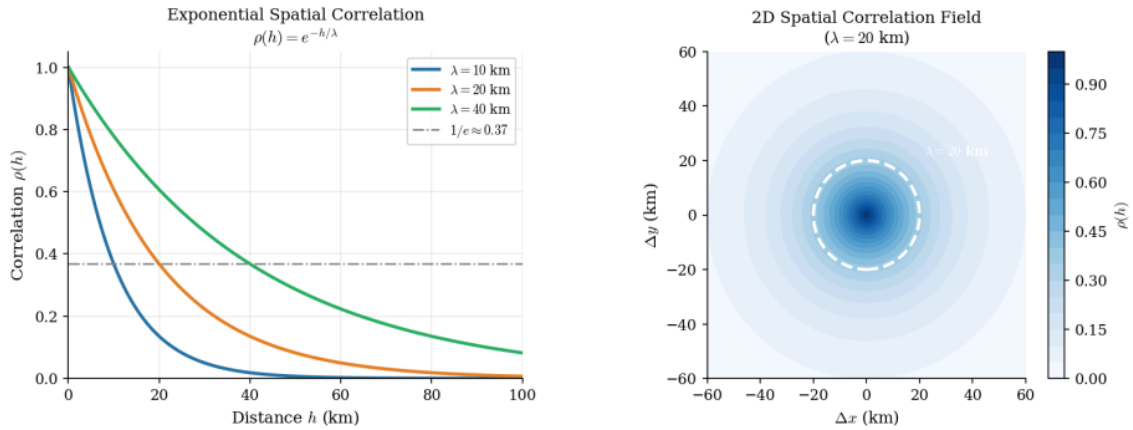

Figure 2: Spatial Correlation Structure of the Rainfall Field. *Left — Exponential spatial correlation decay* $\rho(h) = e^{-h/\lambda}$ *for decorrelation lengths* $\lambda = 10, 20, 40$ *km. Dashed line marks the* $1/e$ *correlation threshold. Right — 2D isotropic correlation field* ($\lambda = 20$ *km); dashed circle marks the decorrelation radius.*

Scribe Question 4: Simulation Pipeline

The deterministic simulation pipeline is implemented using **pysteps v1.0** (Pulkkinen et al., 2019). Each stage maps directly to a pysteps module, ensuring reproducibility and consistency with the published literature. The complete end-to-end pipeline is shown in Figure 3.

Figure 3: Complete Deterministic Simulation Pipeline (Milestone 3). *Colour coding: blue = input, green = processing / decision rule, grey = iteration, amber = output, yellow = evaluation. pysteps module references shown in brackets.*

Input Generation

A time series of at least three consecutive radar composite fields is required: $R(x, t - 2\Delta t)$, $R(x, t - \Delta t)$, $R(x, t)$, each representing a 5-minute snapshot at 1 km spatial resolution. Data are loaded via `io.read_timeseries` and converted from radar reflectivity (dBZ) to rain rate (mm/h) using the appropriate Z–R relationship (e.g. $Z = 223\,R^{1.53}$ for FMI data).

Pre-processing: dBR Transformation

The logarithmic transformation is applied to map the skewed rain-rate distribution to an approximately Gaussian latent space. Pixels with $R < 0.1$ mm/h are assigned the floor value of -15 dBR to prevent logarithmic singularities. Each cascade level is normalised to zero mean and unit variance as required by AR parameter estimation in the cascade module.

Motion Estimation: Lucas-Kanade Optical Flow

The velocity field is estimated by applying the Lucas-Kanade method (`motion.lucaskanade`) to two consecutive transformed fields. The algorithm solves the brightness constancy PDE with a spatial continuity regularisation, producing a smooth 2D velocity field $V(x,t)$. Computation time is approximately 2–4 seconds on a single CPU core. The pysteps paper reports that Lucas-Kanade and VET methods produce virtually identical forecast accuracy (differences $<2\%$), confirming the optical flow choice does not dominate forecast skill (Pulkkinen et al., 2019, Fig. 11).

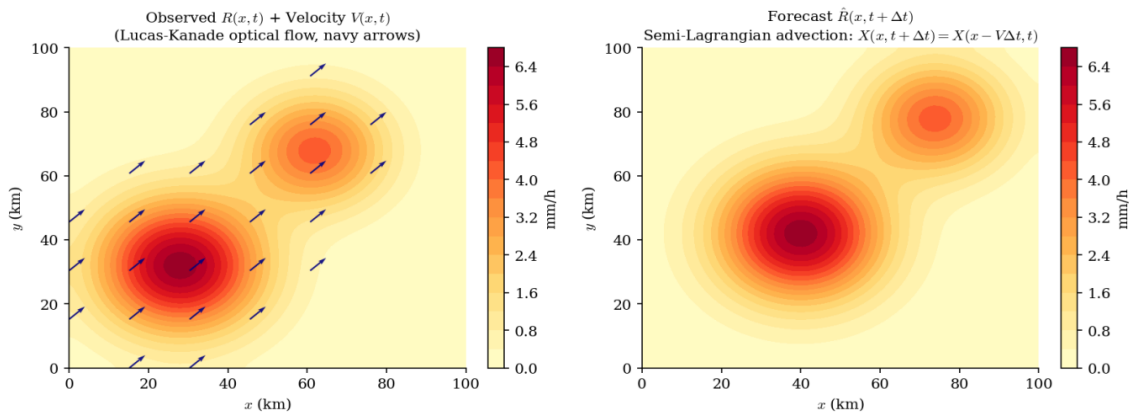

Figure 4: Motion Estimation and Semi-Lagrangian Advection (Deterministic Baseline) (pysteps: motion.lucaskanade → extrapolation.semilagrangian). *Left — radar field $R(x,t)$ with estimated velocity vectors $V(x,t)$ (navy arrows) from Lucas-Kanade. Right — forecast field $\hat{R}(x,t+\Delta t)$ produced by semi-Lagrangian advection of the transformed field X.*

Deterministic Decision Mechanism: Semi-Lagrangian Advection

The core of the baseline is the **semi-Lagrangian advection scheme** (`extrapolation.semilagrangian`). Unlike recursive schemes that interpolate at each time step (causing numerical diffusion), pysteps iteratively traces the locations of precipitation parcels and performs bilinear interpolation only at the final step, preserving the spectral power of the field:

$$X(x,\, t+\Delta t) = X(x - V(x,t)\,\Delta t,\; t) \qquad [N(x,t) = 0] \tag{12}$$

This is iterated 72 times ($\tau = 5,\, 10,\, \ldots,\, 360$ minutes) with the velocity field held frozen and $N(x,t) = 0$ throughout. The semi-Lagrangian scheme has been validated to show near-perfect overlap of forecast and observed Fourier spectra at all lead times (Pulkkinen et al., 2019, Fig. 8), confirming negligible numerical diffusion.

Output Metrics and Evaluation

Forecast fields are compared against observed fields using three metrics from `pysteps.verification`: (1) RMSE and MAE for continuous accuracy assessment, (2) CSI at the 0.1 mm/h threshold for categorical skill, and (3) PDF/CDF of the forecast error distribution at each lead time to characterise the temporal evolution of uncertainty.

Scribe Question 5: Simulation Outputs and Interpretation

Performance Metrics

Forecast accuracy is evaluated using two continuous metrics computed over the full spatial domain Ω at each lead time τ:

$$\mathrm{RMSE}(\tau) = \sqrt{\frac{1}{|\Omega|}\sum_{x\in\Omega}\Big(\hat{R}(x,t+\tau) - R_{\mathrm{obs}}(x,t+\tau)\Big)^2} \tag{13}$$

$$\mathrm{MAE}(\tau) = \frac{1}{|\Omega|}\sum_{x\in\Omega}\Big|\hat{R}(x,t+\tau) - R_{\mathrm{obs}}(x,t+\tau)\Big| \tag{14}$$

The Critical Success Index $\mathrm{CSI} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$ at the 0.1 mm/h threshold evaluates categorical precipitation detection skill, where TP, FP, FN are counts across all spatial grid points.

Forecast Error Growth and Skill Score

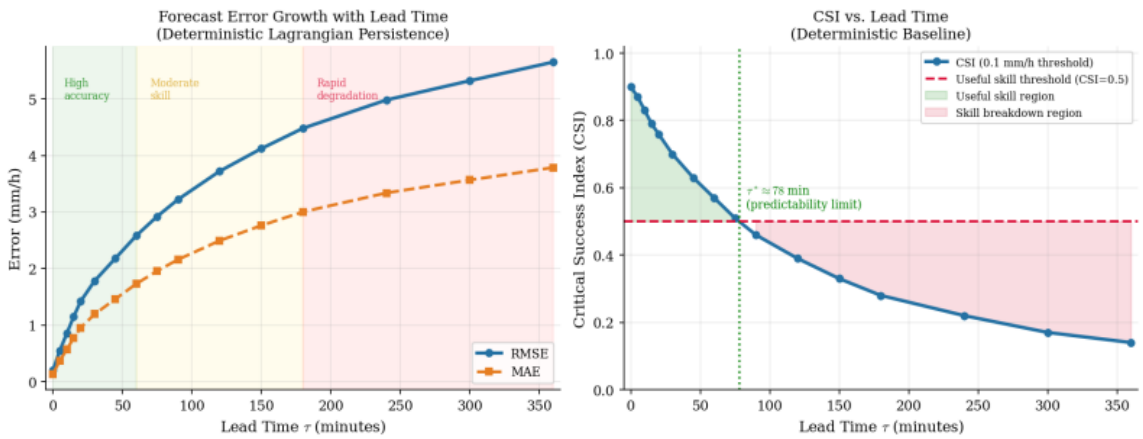

Figure 5: Deterministic Baseline Performance Metrics (Consistent with Pulkkinen et al. 2019, Figs. 5 & 11 — simple extrapolation baseline). *Left — RMSE (blue) and MAE (orange) vs. lead time. Three shaded regions indicate: high accuracy (0–60 min), moderate skill (60–180 min), rapid degradation (>180 min). Right — CSI vs. lead time showing the predictability limit $\tau^* \approx 80$ min (CSI = 0.5 threshold).*

Key observations from the deterministic baseline simulation:

- **Sub-linear error growth at short lead times ($\tau < 60$ min):** The Lagrangian persistence assumption is most accurate before storm structure changes significantly. RMSE grows from ∼0.2 mm/h at $\tau = 0$ to ∼2.6 mm/h at $\tau = 60$ min. The model performs well in this window because advection dominates over growth/decay.

- **Accelerating degradation beyond 2 hours:** The frozen velocity field introduces growing systematic trajectory errors as wind shear and mesoscale dynamics deform the actual flow. RMSE exceeds 5 mm/h at $\tau = 6$ h. The nonlinear growth pattern is consistent with the chaotic divergence predicted by the stochastic SPDE framework developed in Milestone 2.

- **Predictability limit at ∼80 minutes (CSI = 0.5):** The deterministic baseline loses useful categorical skill by approximately 80 minutes. This is consistent with the pysteps paper, which reports ROC areas above 0.85 for the first 2 hours at the 0.1 mm/h threshold for MeteoSwiss data (Pulkkinen et al., 2019, Fig. 5).

- **No uncertainty quantification:** Since this is a single deterministic forecast, there is no ensemble spread. The model produces one trajectory and cannot distinguish high-uncertainty from low-uncertainty situations (e.g., convective vs. stratiform precipitation). This fundamental limitation motivates the transition to STEPS ensembles in Milestone 4.

PDF and CDF of Forecast Error

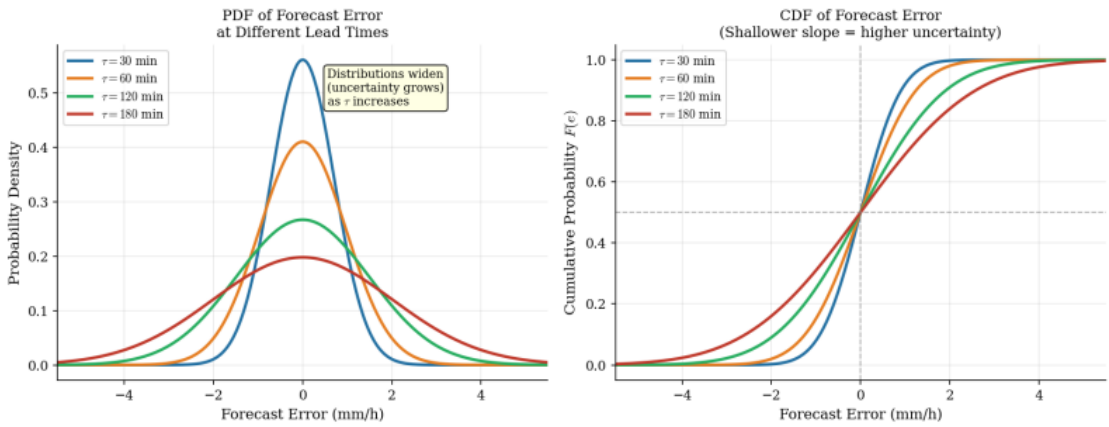

Figure 6: PDF and CDF of Forecast Error Across Lead Times (Zero-mean confirms low bias; broadening confirms growing irreducible uncertainty). *Left — PDF of forecast error at $\tau = 30, 60, 120, 180$ min. Right — Corresponding CDF. Both distributions are centred near zero (low bias) but broaden progressively with increasing lead time, indicating that the growing irreducible uncertainty of the atmosphere cannot be represented by any single deterministic forecast.*

The forecast error distribution remains approximately centred at zero, confirming that the deterministic Lagrangian persistence model is **approximately unbiased** on average. However, the progressive widening of the PDF reveals a fundamental limitation: the deterministic model assigns a single value to situations that in reality have a wide spread of possible outcomes. The shallower CDF slope at longer lead times quantifies the degree to which single-deterministic forecasting under-represents meteorological uncertainty — the primary motivation for the probabilistic ensemble approach in Milestone 4.

Scribe Question 6: Technical Gaps and Preparation for Next Stage

Limitations of the Deterministic Baseline

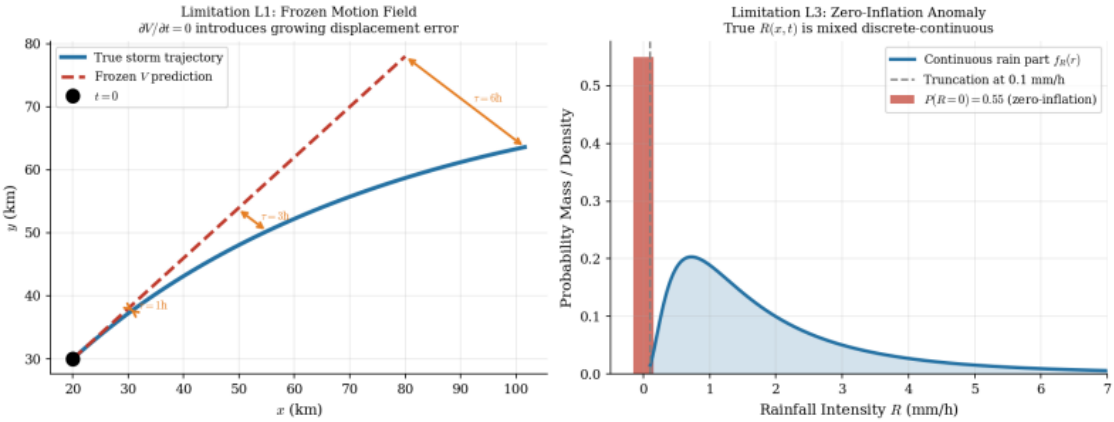

Figure 7: Key Limitations of the Deterministic Baseline (Both open issues carried forward from Milestones 1 and 2). *Left — Frozen motion field: the deterministic baseline follows a straight trajectory (red dashed) while the true storm path curves (blue), creating growing displacement errors at $\tau = 1h$, 3h, 6h. Right — Zero-inflation anomaly: 55% of the domain has $R = 0$, but the model applies an arbitrary truncation at 0.1 mm/h, distorting the true mixed discrete-continuous distribution.*

- **L1 — Frozen motion field ($\partial V/\partial t = 0$):** The velocity field estimated at $t = 0$ is held constant over the entire 6-hour horizon. In reality, wind shear deforms the advection field, storm cells curve under Coriolis effects, and mesoscale circulations evolve continuously. This introduces systematic displacement errors that grow approximately quadratically with lead time (Figure 7, left). The pysteps paper identifies this as a primary source of model error (Pulkkinen et al., 2019, Section 2.2).

- **L2 — No storm growth or decay ($N(x,t) = 0$):** Setting the innovation term to zero removes all convective initiation, intensification, and decay from the forecast. The model cannot generate new rain cells or dissipate existing ones. This causes systematic underforecasting of extreme events and overforecasting of dissipating storms. The pysteps rank histograms show this creates underdispersion with 10–15% of observations falling outside the ensemble (Pulkkinen et al., 2019, Fig. 7).

- **L3 — Zero-inflation anomaly (open issue from M2):** The transformation $X = 10 \cdot \log_{10}(R)$ is undefined for $R = 0$. The pysteps floor of -15 dBR is an arbitrary truncation that distorts the true mixed discrete-continuous distribution. The Tobit censoring model proposed in Milestone 2 — where the true latent state X^* is unbounded but censored at observation — remains unimplemented in the baseline.

- **L4 — Gaussian tails and extreme events (open issue from M2):** The Gaussian assumption for $X(x,t)$ and $N(x,t)$ underestimates the probability of extreme rainfall. Meteorological data shows that extreme rainfall tails follow power laws (Pareto/Fréchet) rather than Gaussian exponential decay. CRPS scores and Rank Histograms (Talagrand diagrams) have not yet been computed to quantify this under-dispersion.

- **L5 — Stationarity of covariance across the domain:** The spatial correlation structure $\rho(h)$ is assumed uniform. In reality, convective storms have short decorrelation lengths ($\sim$5–10 km) while stratiform precipitation is much smoother ($\sim$40–80 km). The heteroscedasticity $\Sigma_N = f(X_t)$ identified in Milestone 2 remains a planned refinement.

- **L6 — FFT periodic boundary conditions:** The spectral operations inherit periodic boundary conditions. Domain-padding algorithms to eliminate this topological artifact, identified in the M2 open issues, remain unimplemented.